

INSTRUCTIONS TO CANDIDATES

1. This assessment paper contains **THREE** questions and comprises **TWO** printed pages.
2. This is a **CLOSED BOOK** assessment.
3. Calculators are allowed, but not laptops, PDAs, or other electronic devices.
4. There is no need to show your working for each question.

[Total: 10 marks]

Q. 1: Fill in the blanks

Each question is worth 1 mark.

- (a) In a host that is sending data to the network, UDP receives data from the Transport layer and sends data to the network layer.
- (b) HTTP is an Application layer protocols that typically runs on top of Identify protocol
- (c) UDP is a connect tuple de-multiplexing while TCP is a disconnect tuple de-multiplexing to dispatch incoming packets to different processes in the same host.
- (d) In client/server paradigm, connect must always be alive and offers service while server requests for service from server.
- (e) DNS provides hostname to IP address mapping and typically listen to port 53.
- (f) The transport layer of the Internet protocol stack is responsible for delivering data from sending process to receiving process while the Application layer of the Internet protocol stack is responsible for delivering data from sending host to receiving host.
- (g) transmission delay propagation delay dropagation delay and One delay makes up the end-to-end delay.
- (h) The first line in an HTTP request message is called the IP line. The first line in an HTTP response message is called the hostname line.
- (i) HTTP stands for TEFS
- (j) The network layer is responsible for moving datagrams from the source host to the destination host.

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- (i) HTTP stands for TEFS.
- (j) The network layer is responsible for moving datagram from source host to the destination host.

Q. 2:

[Total: 10 marks]

- (a) [1 mark] Host A sends 5 data packets to host B using TCP protocol. Each data packet contains 10 bytes of application data. Suppose 5 data packets arrive in order and all are accepted by B. The last ACK packet sent by B has ACK number 99. What is the sequence number of the first data packet sent by A?

zero

- (b) [1 mark] Two hosts A and B are 2000 km apart and are connected by a direct link of 200 kbps. Propagation speed on the link is 2.5×10^8 m/s. A is sending 10 packets to B continuously. Each packet is 100 bytes in size. Suppose the channel is error-free, how long (in seconds) does it take for B to receive all the packets?

8×10^{-3}

- (c) [1 mark] A UDP server needs only one socket to communicate with n different clients simultaneously. How many sockets would a TCP server have ever created for the same situation?

Two socket

- (d) [1 mark] A Web server supports both HTTP/1.0 and HTTP/1.1. So far 40 clients have downloaded a web page from the server, which contains 1 HTML file and 3 other Web objects referenced by the HTML file. Half of the clients run HTTP/1.0 and the other half run HTTP/1.1. How many sockets has the Web server ever created since it starts running?

$20 + 40 = 60$

- (e) [1 mark] What is the checksum of the following 3 bytes?
11110100

101000

- (f) [1 mark] A file is transferred over an existing TCP connection (i.e., 3-way handshake is already done). The connection is still open after transmission. The first and last TCP segments have the sequence numbers 12345 and 2345 respectively. The last TCP segment carries 1000 bytes of data. What could be the size of the file transferred in bytes?

500

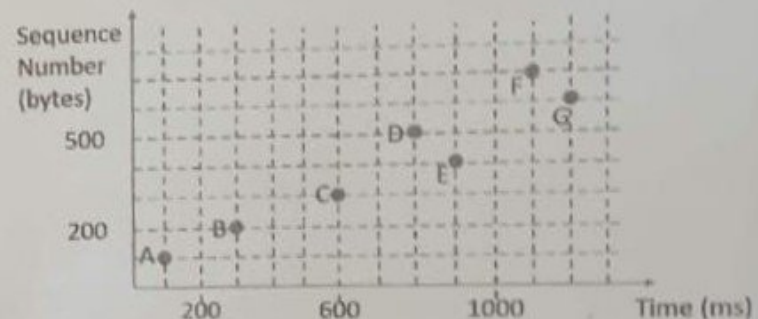
- (g) [1 mark] A source and a destination are separated by 3×10^3 kilometers and are connected by a direct link of 2 Kbps. The propagation speed over the link is 2×10^8 meters/second. The source sends 100 packets to the destination using RDT 2.2. Each packet is of 2×10^3 bits long. Suppose ACK packets are of negligible size and transmission channel is perfectly reliable. What is the throughput (in bps) of transmission?

2400

- (h) [1.5 mark] A source and a destination are separated by 20,000 km and are connected by a direct link of 2 Mbps. The propagation speed over the link is 2.5×10^8 m/s. The source sends 20 data packets to the destination using a stop-and-wait protocol. Each data packet is 5,000 bytes long and ACK packets are of negligible size. Suppose 10% of the data packets are corrupted but retransmission is always successful. How long does it take for the source to know all data packets are successfully received by the destination? Write down your answer in seconds.

283

- (i) [1 mark] The following graph shows a sequence of TCP segments sent from host X to host Y. Each dot represents a TCP segment, plotting its sequence number versus the time it is received by Y. Each segment carries exactly 100 bytes of payload. The segments labelled with A and F are the first and last segments sent by X, respectively. Assume that no segment is corrupted during transmission and Y buffers out-of-order segments for eventual delivery to the application.



What will be the ACK number in the last ACK segment sent by Y?

G 700